BAYESIAN MODELING OF COMPLEXITY

Baysian Reasoning

Meysam Goodarzi
Tuesday 3rd March, 2026

Hertie School of Public Governance

Agenda

1 Recap & Today

2 Introduction

3 Prior, Likelihood, & Posterior

- Prior
- Likelihood
- Posterior
- Conjugate Priors

4 Inference Methods

- Closed-form vs. Sampling

5 Wrap-up & Next

1 Recap & Today

2 Introduction

3 Prior, Likelihood, & Posterior

4 Inference Methods

5 Wrap-up & Next

Last Session Recap

What we covered:

- Random variables, PDF, CDF, joint distributions
- Conditional probability: $P(A \mid B) = \frac{P(A,B)}{P(B)}$
- Bayes' Law: $P(\theta \mid D) = \frac{P(D|\theta)P(\theta)}{P(D)}$
- Medical testing example
- Discrete Bayes update in Python

The fundamental question:

Given observed data **D**, what is the probability of hypothesis θ?

$$P(\theta|\mathbf{D})$$

Today: Deep dive into each component and implement in PyMC

1 Recap & Today

2 Introduction

3 Prior, Likelihood, & Posterior

4 Inference Methods

5 Wrap-up & Next

The Central Question

The fundamental question we would like to answer is:

Considering a phenomenon, what is the probability of hypothesis θ, given observed data **D**?

Mathematically:

$$P(\theta|\mathbf{D})$$

Two schools of thought:

- **Frequentist**
- **Bayesian**

This course focuses on Bayesian methods because they:

- Naturally quantify uncertainty
- Provide interpretable probability statements
- Allow agent-level belief updating

Bayesian vs. Frequentist

Bayesian View

- Hypothesis θ is the source of uncertainty
- θ is **random variable** with a distribution
- Data is **fixed** once observed
- Data updates belief about θ:

$$P(\theta \mid \mathbf{D}) \propto P(\mathbf{D}|\theta)P(\theta)$$

- Credible intervals, direct probability statements

Frequentist View

- Data is random and the source of uncertainty
- Hypothesis θ is **fixed** but **unknown**
- Question: If we repeat experiment, what is the probability of current data?

$$P(\mathbf{D}|\theta)$$

- Inference via long-run frequencies, confidence intervals, p-values

Example: Estimating Coin Bias

Scenario: Observe 7 heads in 10 flips. Estimate $\theta = \text{P(heads)}$

Bayesian Approach:

- θ is random variable
- Prior: $\theta \sim \text{Beta}(2, 2)$ (weakly informative)
- After data: $\theta \mid D \sim \text{Beta}(9, 5)$
- Mean: $\frac{9}{14} \approx 0.64$
- 95% credible interval: [0.39, 0.85]
- **Interpretation:** "95% probability $\theta \in [0.39, 0.85]$"

Frequentist Approach:

- θ is fixed unknown constant
- Maximum likelihood estimate: $\hat{\theta} = \frac{7}{10} = 0.7$
- 95% confidence interval: [0.35, 0.93]
- **Interpretation:** If we repeated this many times, 95% of times, θ will be in the range [0.35, 0.93]
- **Cannot** say "95% probability..."

Key difference: Bayesian gives direct probability statement we intuitively want!

Inference as Probabilistic Reasoning

The central paradigm of probabilistic reasoning is to identify all relevant variables $x_1, \ldots, x_N$ in the environment, and make a probabilistic model $p(x_1, \ldots, x_N)$ of their interactions.

Reasoning (inference) *is then performed by introducing* ***evidence*** *that sets variables in known states, and subsequently computing probabilities of interest, conditioned on this evidence.*

1 Recap & Today

2 Introduction

3 Prior, Likelihood, & Posterior

- Prior
- Likelihood
- Posterior
- Conjugate Priors

4 Inference Methods

5 Wrap-up & Next

Bayes Rule

According to Bayes' rule:

$$P(\theta|\mathbf{D}) = \frac{P(\mathbf{D}|\theta)P(\theta)}{P(\mathbf{D})}$$

Components:

$P(\theta\|\mathbf{D})$:	Posterior	What we learn
$P(\mathbf{D}\|\theta)$:	Likelihood	Fit of data to model
$P(\theta)$:	Prior	Initial belief
$P(\mathbf{D})$:	Evidence	Normalization

Proportional form (often used):

$$P(\theta|\mathbf{D}) \propto P(\mathbf{D}|\theta)P(\theta)$$

The evidence $P(\mathbf{D})$ is constant for given data, so we can ignore it during inference.

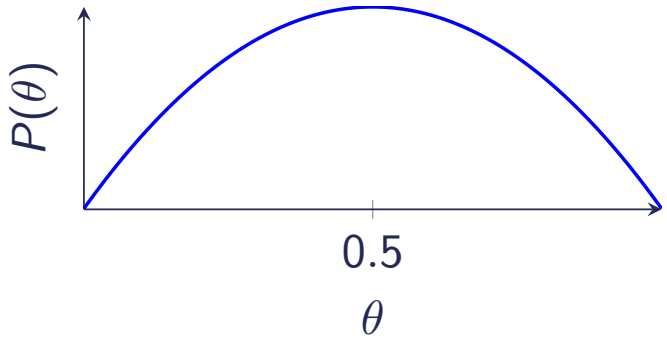
The Prior Distribution

The distribution $P(\theta)$ represents:

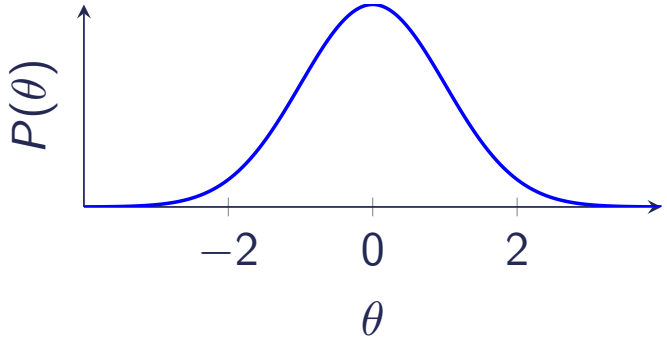
- **Degree of belief** about parameters before seeing current data
- Can be informative or non-informative
- Most controversial aspect of Bayesian inference
- But: Makes assumptions explicit!

Types:

- Uniform: "All values equally likely"
- Weakly informative: Gentle constraints
- Informative: Strong prior knowledge



Beta prior: Beta$(2, 2)$



Normal prior: $\mathcal{N}(0, 1)$

Prior Choice Guidelines

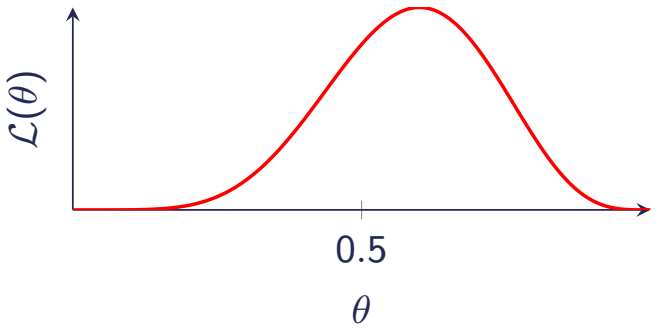
How to choose a prior?

1. **Use domain knowledge**
   - Physical constraints (e.g., probabilities in [0,1])
   - Previous research findings
   - Expert opinions
2. **Start weakly informative**
   - Rule out impossible values
   - Don't strongly constrain plausible values
   - Let data dominate
3. **Check sensitivity**
   - Try different reasonable priors
   - If conclusions robust $\rightarrow$ choice doesn't matter
   - If sensitive $\rightarrow$ need more data or better prior

The Likelihood Function

Definition: $P(\mathbf{D} \mid \theta)$

- Probability of observed data as function of parameters
- Comes from data-generating assumption
- Examples: Binomial, Poisson, Normal, etc.
- Drives how strongly data pulls on prior



Binomial likelihood: 6 successes in 10 trials. Peak at $\theta = 0.6$ (maximum likelihood estimate).

Key insight:

- As function of D: probability distribution
- As function of θ: likelihood (**does not sum to 1!**)

Writing Likelihood Functions

Recipe:

1. Choose probability distribution for each data point
2. Identify parameters
3. Write probability of each observation
4. Multiply (if independent)

Example: Heights

- Data: $y_1, \ldots, y_n$
- Model: $y_i \sim \mathcal{N}(\mu, \sigma^2)$ independently
- Likelihood: $P(D \mid \mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y_i-\mu)^2}{2\sigma^2}\right)$

Example: Coin flips

- Data: k heads in n trials
- Model: Each flip $\sim \text{Bernoulli}(\theta)$
- Likelihood: $P(k \mid \theta, n) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$

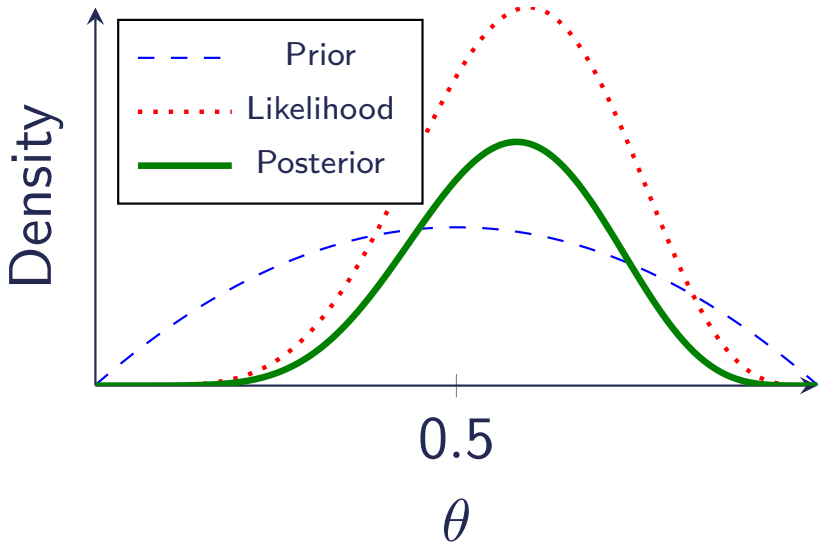
The Posterior Distribution

We recall the the posterior is written as:

$$p(\theta \mid \mathbf{D}) = \frac{p(\mathbf{D} \mid \theta)\, p(\theta)}{p(\mathbf{D})}$$

where

$$p(\mathbf{D}) = \int p(\mathbf{D} \mid \theta) p(\theta)\, d\theta$$



Compromise between prior and likelihood

Properties:

- Balances prior information and data evidence
- Quantifies uncertainty (intervals, predictions)
- Converges to truth as $n \to \infty$
- Today's posterior = tomorrow's prior

Posterior as a Compromise

Example: Coin with 6 heads in 10 flips

- **Prior:** Beta(2,2), peak at 0.5
- **Likelihood:** Peak at 0.6 (MLE)
- **Posterior:** Beta(8,6), peak at $\frac{8-1}{8+6-2} \approx 0.58$

Observation: Posterior is between prior and likelihood!

General pattern:

- Little data $\rightarrow$ posterior close to prior
- Lots of data $\rightarrow$ posterior close to MLE
- Always some compromise

As $n \rightarrow \infty$:

- Posterior concentrates around true value
- Uncertainty decreases
- Prior becomes negligible

Conjugate Families

Definition: Prior is **conjugate** if the posterior has the same form

- Posterior $P(\theta|\mathbf{D})$ may not have same distribution as prior $P(\theta)$
- Conjugacy simplifies computation
- Analytical solutions possible
- Clear interpretation (add counts)

Importance today:

- Conjugates **used to be** crucial (no computers)
- Now: MCMC handles non-conjugate (next session!)
- Still useful for intuition and simple models

Common conjugate pairs:

Prior	**Likelihood**
Beta	Binomial
Normal	Normal
Gamma	Poisson
Gamma	Exponential

Example I: Beta-Binomial (Conjugate)

Setup:

- Prior: $\theta \sim \text{Beta}(\alpha, \beta)$
- Data: k successes in n trials
- Likelihood: $\text{Binomial}(n, \theta)$

Result (conjugacy):

Posterior:

$$\theta \mid D \sim \text{Beta}(\alpha + k, \beta + n - k)$$

Interpretation:

- Start: α pseudo-successes, β pseudo-failures
- Observe: k actual successes, $n - k$ actual failures
- Update: Add observed to pseudo-counts

Concrete example:

- Prior: Beta(2, 2)
- Data: 6 heads in 10 flips
- Posterior: Beta(2+6, 2+4) = Beta(8, 6), Mean: $\frac{8}{14} \approx 0.57$

Example II: Normal-Normal (Conjugate)

Setup: Estimate mean μ with known variance σ^2

Prior: $\mu \sim \mathcal{N}(\mu_0, \tau_0^2)$

Data: $y_1, \ldots, y_n \sim \mathcal{N}(\mu, \sigma^2)$

Posterior (conjugate):

$$\mu \mid D \sim \mathcal{N}(\mu_n, \tau_n^2)$$

where

$$\mu_n = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}, \quad \tau_n^2 = \frac{1}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}$$

Intuition: Posterior mean is weighted average of prior mean and data mean, weighted by precisions (inverse variances)

Example III: Powerlaw (Non-conjugate)

Powerlaw distributions appear in many complex systems:

- City sizes
- Income distributions
- Network degree distributions
- ABM outputs (sometimes)

Model: $P(x) \propto x^{-\alpha}$ for $x \geq x_{\text{min}}$

Parameter: α (exponent)

Likelihood: $P(D \mid \alpha) = \prod_{i=1}^{n} \frac{\alpha-1}{x_{\text{min}}} \left(\frac{x_i}{x_{\text{min}}}\right)^{-\alpha}$

Challenge: No standard conjugate prior for α!

Solution: ???

1 Recap & Today

2 Introduction

3 Prior, Likelihood, & Posterior

4 Inference Methods

- Closed-form vs. Sampling

5 Wrap-up & Next

Closed-Form vs. Sampling

Analytical (Closed-form):

- **Conjugate** models → analytic posteriors
- Fast, exact, transparent
- Limited to simple models
- Examples: Beta-Binomial, Normal-Normal

Computational (Sampling):

- Complex models → MCMC or Variational Inference
- Approximate but flexible
- Handles hierarchical, nonlinear models

MCMC Chain (preview):

Markov chain samples

Key issues:

- Convergence
- Mixing
- Effective sample size
- Diagnostics ($\hat{R}$, ESS)
- **Next session:** MCMC in detail

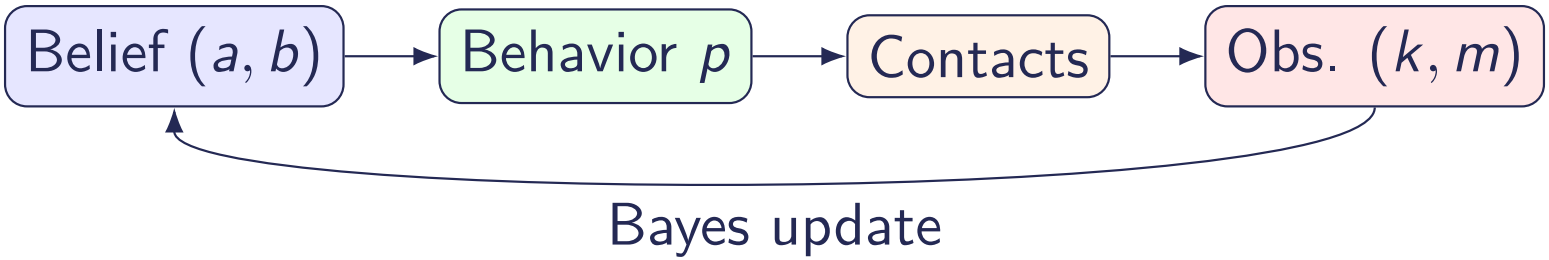
Bayesian Agents

Bayesian agents: Maintain beliefs, update with Bayes' Law

Example: Epidemic with belief updating

Each agent:

- Holds Beta prior over disease prevalence θ: $\text{Beta}(a, b)$
- Observes k infected among m contacts
- Updates: $(a, b) \leftarrow (a + k, b + m - k)$
- Protection decision based on posterior mean: $p = \sigma(\alpha(\mathbb{E}[\theta] - c))$
- Behavior feeds back to contacts



1 Recap & Today

2 Introduction

3 Prior, Likelihood, & Posterior

4 Inference Methods

5 Wrap-up & Next

Summary

Key concepts covered:

- **Prior:** Encodes initial beliefs
- **Likelihood:** Connects parameters to data
- **Posterior:** Updated beliefs after seeing data
- **Conjugacy:** Analytical updates (when available)
- **PyMC:** Computational Bayesian inference

Bayesian vs. Frequentist:

- Both valid frameworks
- Bayesian: Direct probability statements, natural for ABMs
- Frequentist: No priors needed, well-established

Applications to ABMs:

- Parameter calibration
- Agent belief updating
- Quantification of uncertainty

Homework

Before the next session:

1. **Practice Bayesian Reasoning:**
   - Work through examples in the book
2. **PyMC exercises:**
   - Implement simple priors and likelihoods
   - Implement sequential updating
   - Check the sensitivity to prior
   - Compare analytical results with PyMC
   - Visualize your steps
3. **Reading:**
   - Gill Ch. 2 (Bayes' Theorem)
4. **Play around PyMC**, e.g, go through CoVID example.

Next Session: Session 6

MCMC & PyMC Deep Dive

Topics:

- Why MCMC? (when conjugacy fails)
- Metropolis-Hastings algorithm
- Gibbs sampling
- PyMC sampling internals
- **Critical:** MCMC diagnostics
  - Trace plots
  - $\hat{R}$ (Gelman-Rubin)
  - Effective sample size
  - Divergences
- Posterior predictive checks

Then: Midterm exam (Sessions 1-6)

Questions?

Next: MCMC